Practice workbook · Where lines cross: two routes and comparing coefficients

Finding the position of a point where two lines cross, or a point fixed by a collinear or parallel condition, by writing its vector in two ways with unknown scalars and comparing coefficients of the two non-parallel base vectors.

20 question parts: 1 worked-example retry and 19 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-VEC-07

1. Question 11 · part b · 7 marks

2023 JAN · Paper 01

Worked-example retry · Attempt part b only.

11

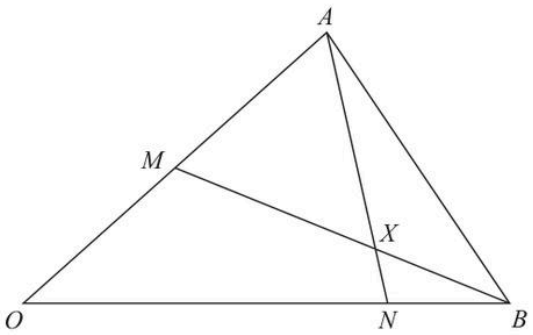


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows triangle OAB with $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OB} = \mathbf{b}$

M is the midpoint of OA.

N is the point on OB such that $ON : NB = 3 : 1$

The lines AN and BM intersect at the point X.

(a) Find expressions, in terms of **a** and **b**, for

(i) $\overrightarrow{AN}$ (ii) $\overrightarrow{BM}$

(3)

(b) Using a vector method, find $AX : XN$

(7)

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1. Question 11 · part b · 7 marks

2023 JAN · Paper 01

Worked-example retry · Attempt part b only.

Question 11 continued

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1. Question 11 · part b · 7 marks

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Worked-example retry · Attempt part b only.

Question 11 continued

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(Total for Question 11 is 10 marks)

1. Question 11 · part b · 7 marks · Working and answer

Working and answer

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2. Question 11 · part b · 4 marks

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Further practice · Attempt part b only.

11

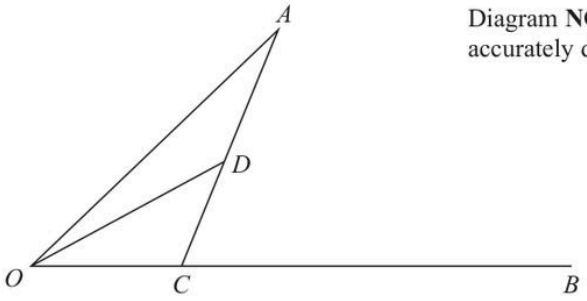


Diagram **NOT** accurately drawn

Figure 5

In Figure 5, $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OB} = \mathbf{b}$

The point C divides OB in the ratio $1 : 3$
The point D is the midpoint of AC

(a) Find, as a simplified expression in terms of **a** and **b**

(i) $\overrightarrow{AC}$ (ii) $\overrightarrow{OD}$ (iii) $\overrightarrow{BD}$

(5)

The point E is such that $\overrightarrow{OE} = \lambda\overrightarrow{OA}$

Given that E, D and B are collinear

(b) find the value of λ

(4)

Given that $\dfrac{\text{area } \Delta OAC}{\text{area } \Delta OEB} = \mu$

(c) find the value of μ

(4)

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P 6 2 2 8 0 A 0 3 0 3 2

2. Question 11 · part b · 4 marks · Working and answer

Working and answer

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3. Question 11 · part b · 6 marks

2023 NOV · Paper 02

Further practice · Attempt part b only.

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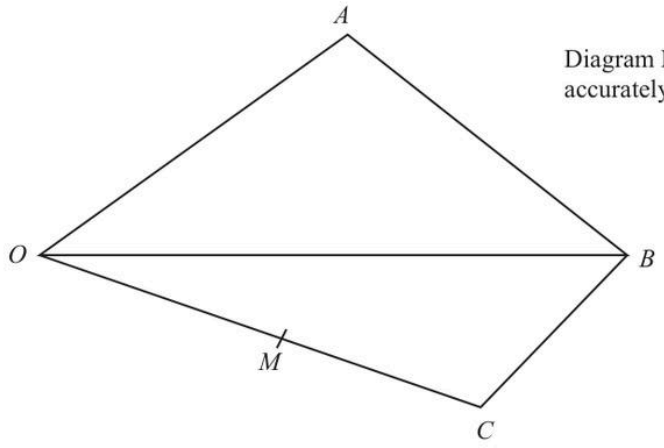


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows quadrilateral $OABC$ where

$$\overrightarrow{OA} = 4\mathbf{p} + 5\mathbf{q} \quad \overrightarrow{OB} = 3\mathbf{p} + \mathbf{q} \quad \overrightarrow{OC} = 2\mathbf{p} - 4\mathbf{q}$$

The point M is the midpoint of OC

(a) Find $\overrightarrow{MA}$ as a simplified expression in terms of $\mathbf{p}$ and $\mathbf{q}$

(3)

The point N lies on OB such that M, N and A are collinear.

(b) Find the ratio $MN : NA$

(6)

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3. Question 11 · part b · 6 marks

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Further practice · Attempt part b only.

Question 11 continued

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3. Question 11 · part b · 6 marks

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Further practice · Attempt part b only.

Question 11 continued

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(Total for Question 11 is 9 marks)

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3. Question 11 · part b · 6 marks · Working and answer

Working and answer

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4. Question 11 · part a · 9 marks

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part a only.

11

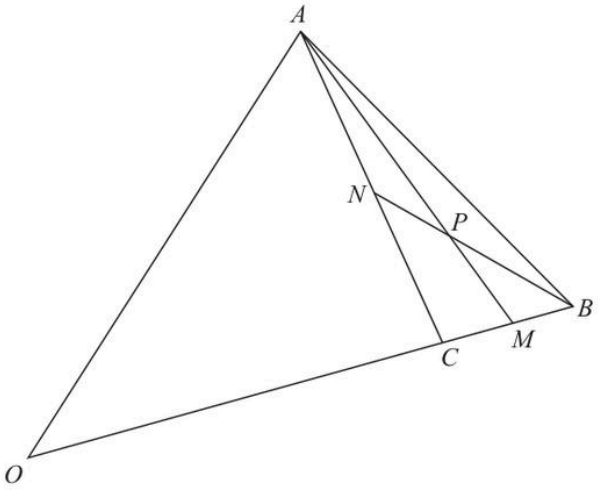


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows triangle OAB with $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OB} = \mathbf{b}$

The point C lies on OB such that $OC : CB = 2 : 1$
The point M is the midpoint of CB and the point N is the midpoint of AC
The lines AM and NB intersect at the point P

(a) Using a vector method, find $\overrightarrow{OP}$ as a simplified expression in terms of **a** and **b**

(9)

The point Q is the midpoint of AB

(b) Using a vector method, show that C, P and Q are collinear.

(4)

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4. Question 11 · part a · 9 marks · Working and answer

Working and answer

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5. Question 8 · part b · 5 marks

2024 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

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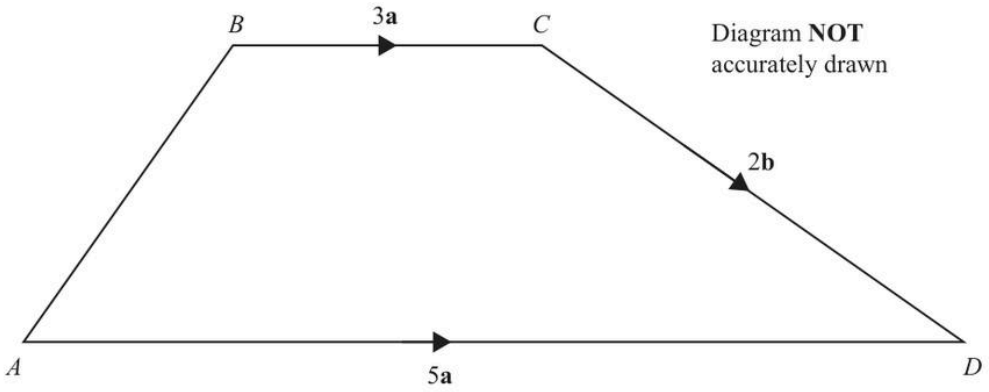


Figure 3

Figure 3 shows a trapezium $ABCD$

$\overrightarrow{BC} = 3\mathbf{a}$ $\overrightarrow{AD} = 5\mathbf{a}$ $\overrightarrow{CD} = 2\mathbf{b}$

(a) Find $\overrightarrow{AB}$ as a simplified expression in terms of **a** and **b**

(1)

The diagonals BD and AC intersect at point X where $\overrightarrow{BX} = k\overrightarrow{BD}$

(b) Using a vector method, find the value of k

(5)

(c) Find the ratio of the area of triangle CXD : area of the trapezium $ABCD$

(4)

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5. Question 8 · part b · 5 marks · Working and answer

Working and answer

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6. Question 9 · part d · 6 marks

2022 JAN · Paper 02

Further practice · Attempt part d only.

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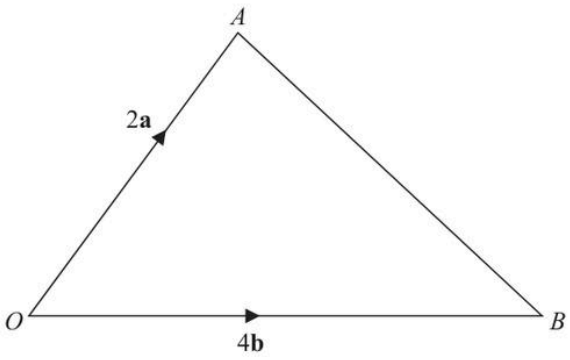


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Figure 3

Figure 3 shows the triangle OAB with

$$\overrightarrow{OA} = 2\mathbf{a} \text{ and } \overrightarrow{OB} = 4\mathbf{b}$$

(a) Find $\overrightarrow{AB}$ in terms of **a** and **b**

(2)

The point P is the midpoint of AB

(b) Find $\overrightarrow{OP}$ as a simplified expression in terms of **a** and **b**

(2)

The point Q lies on OP such that $OQ : QP = 3 : 1$

(c) Find $\overrightarrow{AQ}$ as a simplified expression in terms of **a** and **b**

(3)

The point R lies on OB such that AQR is a straight line.

(d) Find in its simplest form the ratio $OR : RB$

(6)

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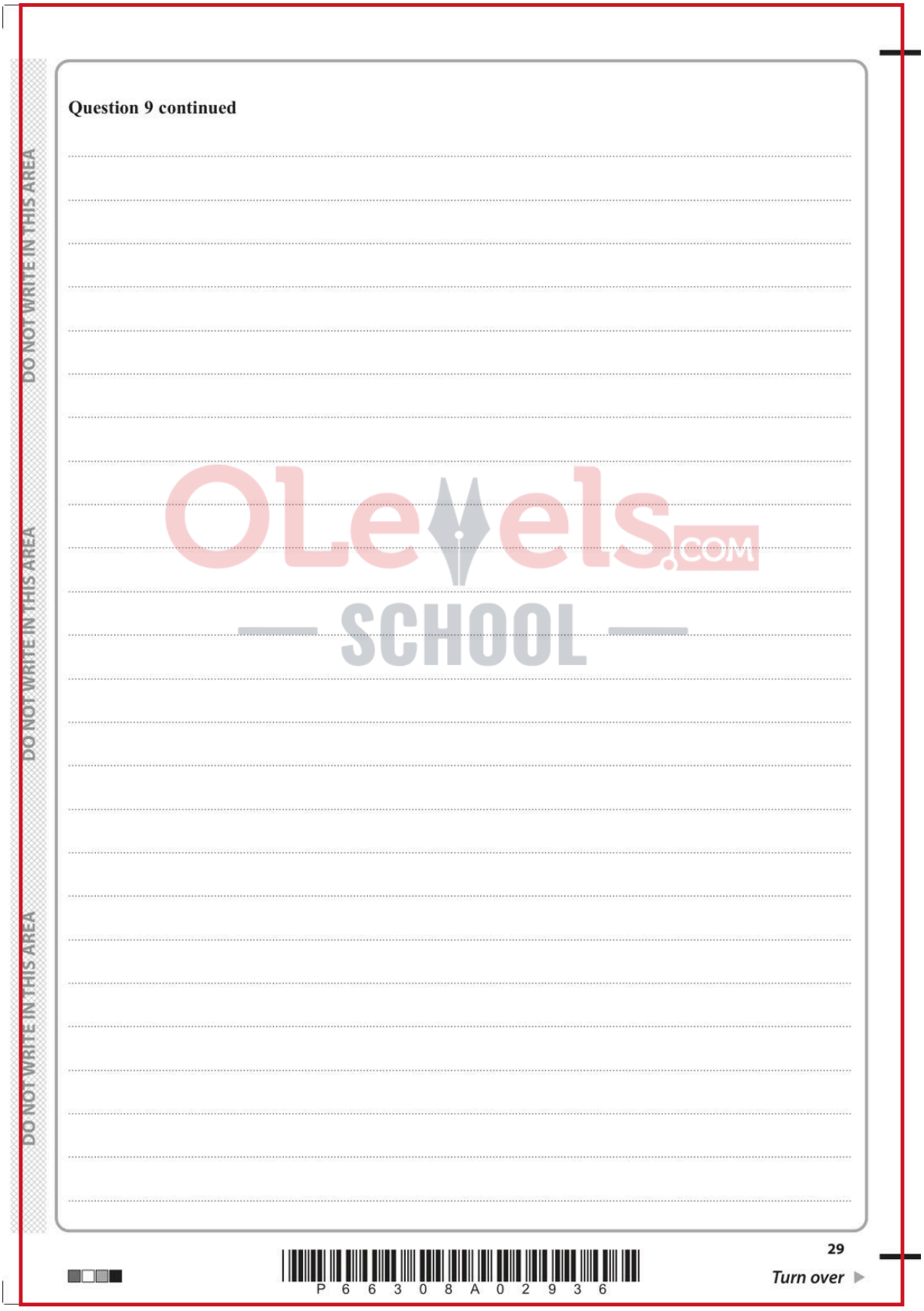
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6. Question 9 · part d · 6 marks

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Further practice · Attempt part d only.



Question 9 continued

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Turn over

6. Question 9 · part d · 6 marks

2022 JAN · Paper 02

Further practice · Attempt part d only.



Question 9 continued

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6. Question 9 · part d · 6 marks

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Further practice · Attempt part d only.

Question 9 continued

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(Total for Question 9 is 13 marks)

6. Question 9 · part d · 6 marks · Working and answer

Working and answer

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7. Question 8 · part b · 4 marks

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Further practice · Attempt part b only.

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Diagram **NOT** accurately drawn

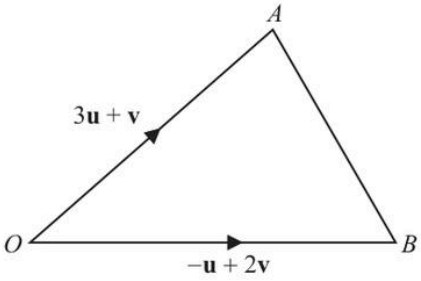


Figure 2

Figure 2 shows triangle OAB with

$$\overrightarrow{OA} = 3\mathbf{u} + \mathbf{v} \qquad \overrightarrow{OB} = -\mathbf{u} + 2\mathbf{v}$$

(a) Find $\overrightarrow{AB}$ as a simplified expression in terms of $\mathbf{u}$ and $\mathbf{v}$ (2)

Point C is such that AC is parallel to OB

Given that $\overrightarrow{OC} = \mu\mathbf{v}$

(b) find the value of μ (4)

The lines OC and AB intersect at point X

Point D lies on $\overrightarrow{OC}$ such that

area of triangle BOX : area of triangle $BXD = 2 : 3$

Using a vector method,

(c) find $\overrightarrow{BD}$ as a simplified expression in terms of $\mathbf{u}$ and $\mathbf{v}$ (5)

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7. Question 8 · part b · 4 marks · Working and answer

Working and answer

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8. Question 10 · part c · 7 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part c only.

10

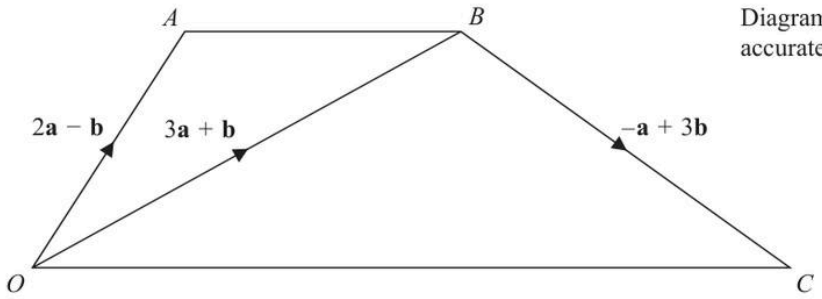


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows quadrilateral $OABC$ with

$$\overrightarrow{OA} = 2\mathbf{a} - \mathbf{b} \qquad \overrightarrow{OB} = 3\mathbf{a} + \mathbf{b} \qquad \overrightarrow{BC} = -\mathbf{a} + 3\mathbf{b}$$

(a) Find $\overrightarrow{AB}$ as a simplified expression in terms of **a** and **b**.

(2)

(b) Prove that $\overrightarrow{OC}$ is parallel to $\overrightarrow{AB}$

(2)

The diagonals, OB and AC, intersect at the point X.

(c) Using a vector method find the ratio $AX : XC$

(7)

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8. Question 10 · part c · 7 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 10 continued

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8. Question 10 · part c · 7 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 10 continued

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8. Question 10 · part c · 7 marks

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Further practice · Attempt part c only.

Question 10 continued

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(Total for Question 10 is 11 marks)

8. Question 10 · part c · 7 marks · Working and answer

Working and answer

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9. Question 10 · part b · 7 marks

2021 JUN · Paper 01

Further practice · Attempt part b only.

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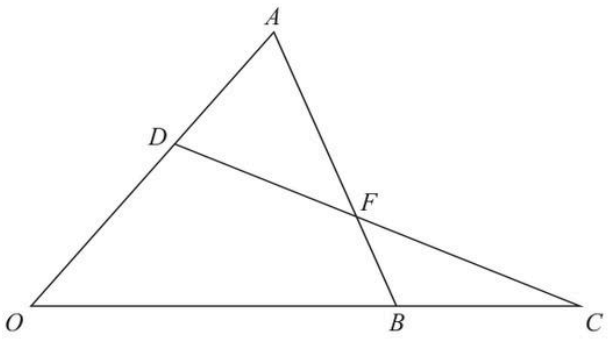


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows triangle OAB and triangle OCD.

$$\overrightarrow{OA} = 5\mathbf{p} \quad \overrightarrow{AB} = 3\mathbf{q} \quad \overrightarrow{OC} = \frac{3}{2}\overrightarrow{OB} \quad \overrightarrow{OD} = \frac{3}{5}\overrightarrow{OA}$$

(a) Find $\overrightarrow{DC}$ as a simplified expression in terms of $\mathbf{p}$ and $\mathbf{q}$.

(3)

The line DC meets the line AB at F.

(b) Using a vector method, find $\overrightarrow{OF}$ as a simplified expression in terms of $\mathbf{p}$ and $\mathbf{q}$.

(7)

The point G lies on OB such that FG is parallel to AO.

(c) Using a vector method, find $\overrightarrow{OG}$ as a simplified expression in terms of $\mathbf{p}$ and $\mathbf{q}$.

(4)

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9. Question 10 · part b · 7 marks · Working and answer

Working and answer

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10. Question 10 · part c · 4 marks

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Further practice · Attempt part c only.

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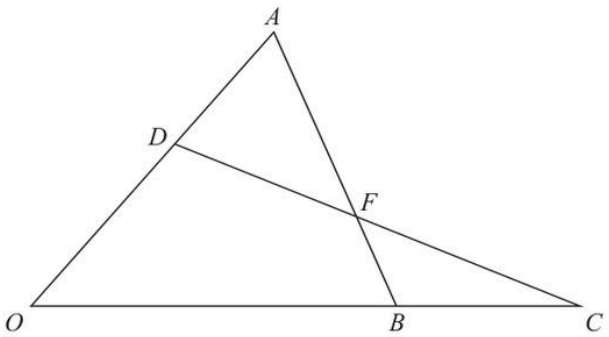


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows triangle OAB and triangle OCD.

$$\overrightarrow{OA} = 5\mathbf{p} \quad \overrightarrow{AB} = 3\mathbf{q} \quad \overrightarrow{OC} = \frac{3}{2}\overrightarrow{OB} \quad \overrightarrow{OD} = \frac{3}{5}\overrightarrow{OA}$$

(a) Find $\overrightarrow{DC}$ as a simplified expression in terms of **p** and **q**.

(3)

The line DC meets the line AB at F.

(b) Using a vector method, find $\overrightarrow{OF}$ as a simplified expression in terms of **p** and **q**.

(7)

The point G lies on OB such that FG is parallel to AO.

(c) Using a vector method, find $\overrightarrow{OG}$ as a simplified expression in terms of **p** and **q**.

(4)

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10. Question 10 · part c · 4 marks

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Further practice · Attempt part c only.

Question 10 continued

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10. Question 10 · part c · 4 marks

2021 JUN · Paper 01

Further practice · Attempt part c only.

Question 10 continued

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10. Question 10 · part c · 4 marks

2021 JUN · Paper 01

Further practice · Attempt part c only.

Question 10 continued

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(Total for Question 10 is 14 marks)

10. Question 10 · part c · 4 marks · Working and answer

Working and answer

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11. Question 9 · part b · 6 marks

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Further practice · Attempt part b only.

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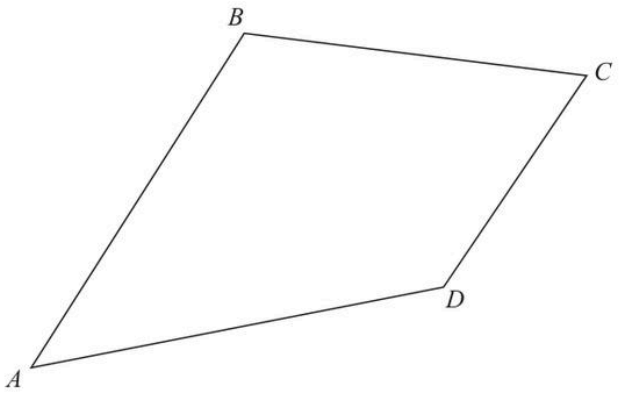


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows quadrilateral $ABCD$ such that

$$\overrightarrow{AD} = 2\mathbf{a} + \mathbf{b} \qquad \overrightarrow{BC} = \frac{1}{3}\mathbf{b} \qquad \overrightarrow{BD} = -4\mathbf{a} - \mathbf{b}$$

(a) Prove that $\overrightarrow{AB}$ is parallel to $\overrightarrow{DC}$ **(4)**

The diagonals, AC and BD, of the quadrilateral intersect at the point Y.

(b) Using a vector method, find $\overrightarrow{AY}$ as a simplified expression in terms of **a** and **b** **(6)**

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11. Question 9 · part b · 6 marks

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Further practice · Attempt part b only.

Question 9 continued

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11. Question 9 · part b · 6 marks

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Further practice · Attempt part b only.

Question 9 continued

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11. Question 9 · part b · 6 marks

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

Question 9 continued

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(Total for Question 9 is 10 marks)

11. Question 9 · part b · 6 marks · Working and answer

Working and answer

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12. Question 10 · part b · 5 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

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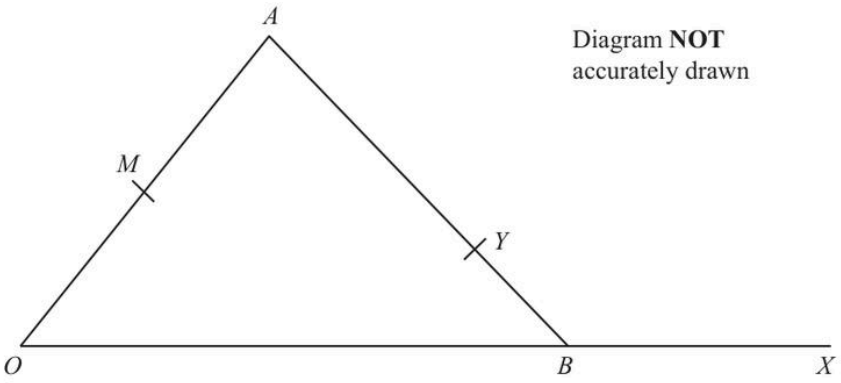


Diagram **NOT** accurately drawn

Figure 5

In Figure 5, $\overrightarrow{OA} = 2\mathbf{a}$, $\overrightarrow{OB} = 4\mathbf{b}$ and M is the midpoint of OA.

The point Y lies on AB such that $AY : YB = 3 : 1$

The point X lies on OB produced.

(a) Find as simplified expressions in terms of **a** and **b**

(i) $\overrightarrow{AB}$ (ii) $\overrightarrow{MY}$

(3)

The points M, Y and X are collinear.

(b) Find the ratio $OB : OX$

(5)

(c) Find the ratio of $(\text{Area } \Delta YBX) : (\text{Area } \Delta OAX)$

(3)

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12. Question 10 · part b · 5 marks · Working and answer

Working and answer

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13. Question 10 · part b · 8 marks

2020 JAN · Paper 01

Further practice · Attempt part b only.

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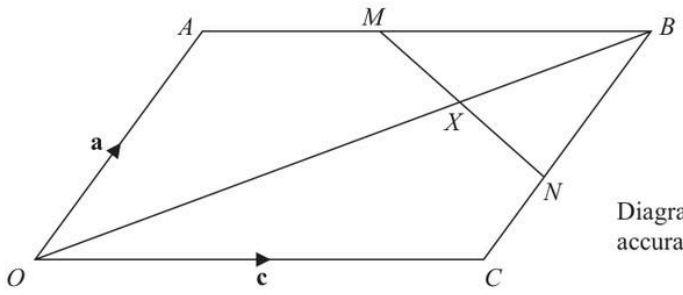


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows the parallelogram $OABC$

$$\overrightarrow{OA} = \mathbf{a} \qquad \overrightarrow{OC} = \mathbf{c}$$

The midpoint of AB is M and the midpoint of BC is N.

The line OB intersects MN at the point X.

(a) Find in terms of **a** and **c**,

(i) $\overrightarrow{OB}$

(ii) $\overrightarrow{MN}$

(2)

Given $\overrightarrow{MX} = \lambda\overrightarrow{MN}$ and that $\overrightarrow{OX} = \mu\overrightarrow{OB}$,

(b) use a vector method to find the value of λ and the value of μ.

(8)

(c) Hence find, in its simplest form, the ratio

Area of quadrilateral $OXNC$: Area of parallelogram $OABC$.

(3)

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13. Question 10 · part b · 8 marks · Working and answer

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14. Question 10 · part b · 6 marks

2022 JUN · Paper 01

Further practice · Attempt part b only.

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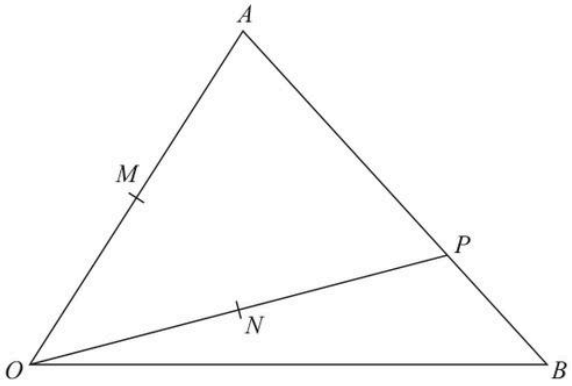


Diagram **NOT** accurately drawn

Figure 4

Figure 4 shows triangle OAB in which

$$\overrightarrow{OA} = \mathbf{a} \text{ and } \overrightarrow{OB} = \mathbf{b}$$

The point P lies on AB such that $AP : PB = 3 : 1$

The point M is the midpoint of OA and the point N is the midpoint of OP.

(a) Find, as simplified expressions in terms of **a** and **b**, the vector

(i) $\overrightarrow{OP}$ (ii) $\overrightarrow{MN}$

(4)

The point C lies on OB such that ANC is a straight line.

(b) Using a vector method, find the vector $\overrightarrow{OC}$ as a simplified expression in terms of **b**

(6)

Given that $\dfrac{\text{area of quadrilateral } AMNP}{\text{area of triangle } OAB} = K$

(c) find the exact value of K

(4)

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14. Question 10 · part b · 6 marks · Working and answer

Working and answer

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15. Question 10 · part b · 8 marks

2024 JUN · Paper 02

Further practice · Attempt part b only.

10 The points A, B, C and D are the vertices of a quadrilateral such that

$$\overrightarrow{AB} = 3\mathbf{a} + 4\mathbf{b} \quad \overrightarrow{AC} = 7\mathbf{a} + 9\mathbf{b} \quad \overrightarrow{AD} = 4\mathbf{a} + 5\mathbf{b}$$

(a) Show that $ABCD$ is a parallelogram.

(3)

BC is extended to the point E such that BCE is a straight line.

Point F lies on CD such that $CF : FD = 1 : 2$

Given that A, F and E are collinear,

(b) find the vector $\overrightarrow{AE}$ in the form $X\mathbf{a} + Y\mathbf{b}$ where X and Y are rational numbers to be found.

(8)

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15. Question 10 · part b · 8 marks

2024 JUN · Paper 02

Further practice · Attempt part b only.

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Question 10 continued

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P 7 6 5 0 8 A 0 2 7 3 2

15. Question 10 · part b · 8 marks

2024 JUN · Paper 02

Further practice · Attempt part b only.

Question 10 continued

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15. Question 10 · part b · 8 marks

2024 JUN · Paper 02

Further practice · Attempt part b only.

Question 10 continued

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(Total for Question 10 is 11 marks)

15. Question 10 · part b · 8 marks · Working and answer

Working and answer

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16. Question 9 · part c.i · 4 marks

2025 NOV · Paper 02

Further practice · Attempt part c.i only.

9 The points A, B, C and D are the vertices of a quadrilateral where

$$\overrightarrow{AB} = 2\mathbf{a} + 4\mathbf{b} \qquad \overrightarrow{AC} = 7\mathbf{a} + 7\mathbf{b} \qquad \overrightarrow{AD} = 10\mathbf{a} + 6\mathbf{b}$$

(a) Show that $ABCD$ is a trapezium.

(4)

Given that $|\mathbf{a}| = 1$ and $|\mathbf{b}| = 1$ and that $\mathbf{a}$ and $\mathbf{b}$ are perpendicular to each other

(b) find a unit vector parallel to $\overrightarrow{BD}$

(4)

The point Y lies on CD such that $CY : YD = 1 : 2$

The lines AC and BY intersect at the point X

(c) (i) Use a vector method to find the ratio $BX : XY$

(4)

(ii) Hence find, in its simplest form, the ratio

area of triangle CXY : area of trapezium $ABCD$

(4)

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16. Question 9 · part c.i · 4 marks · Working and answer

Working and answer

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17. Question 8 · part d · 5 marks

2021 NOV · Paper 01

Further practice · Attempt part d only.

8

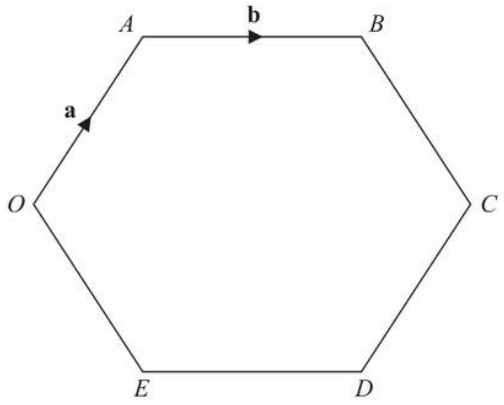


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows the regular hexagon $OABCDE$ with $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{AB} = \mathbf{b}$

(a) Find $\overrightarrow{OB}$ in terms of **a** and **b**

(1)

(b) Find $\overrightarrow{BC}$ as a simplified expression in terms of **a** and **b**

(3)

The point M divides BC in the ratio $2:1$

(c) Find $\overrightarrow{OM}$ as a simplified expression in terms of **a** and **b**

(2)

The point Y is such that OMY and ABY are straight lines.

(d) Use a vector method to find $AB:BY$

(5)

The area of hexagon $OABCDE$ is $60\,\text{cm}^2$

(e) Find the area of triangle OAY

(4)

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17. Question 8 · part d · 5 marks · Working and answer

Working and answer

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18. Question 8 · part c · 5 marks

2024 NOV · Paper 02

Further practice · Attempt part c only.

8

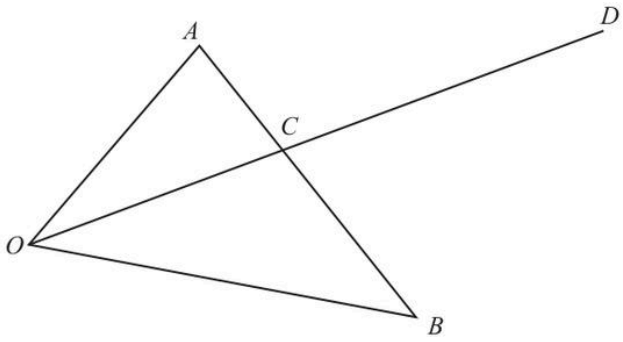


Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows triangle AOB

$\overrightarrow{OA} = 4\mathbf{a} + 5\mathbf{b}$ $\overrightarrow{OB} = 8\mathbf{a} - \mathbf{b}$ $\overrightarrow{OD} = 15\mathbf{a} + 10\mathbf{b}$ where $|\mathbf{a}| = |\mathbf{b}| = 1$

(a) (i) Find $\overrightarrow{AB}$ in terms of $\mathbf{a}$ and $\mathbf{b}$

(ii) Find, in its simplest form, the exact value of $|\overrightarrow{AB}|$ **(3)**

(b) Find the area of triangle AOB **(4)**

The point C lies on AB and OD such that O, C and D are collinear.

(c) Use a vector method to find vector $\overrightarrow{OC}$ as a simplified expression in terms of $\mathbf{a}$ and $\mathbf{b}$ **(5)**

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18. Question 8 · part c · 5 marks

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Further practice · Attempt part c only.

Question 8 continued

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18. Question 8 · part c · 5 marks

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Further practice · Attempt part c only.

Question 8 continued

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18. Question 8 · part c · 5 marks

2024 NOV · Paper 02

Further practice · Attempt part c only.

Question 8 continued

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(Total for Question 8 is 12 marks)

18. Question 8 · part c · 5 marks · Working and answer

Working and answer

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19. Question 11 · part b · 9 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part b only.

11

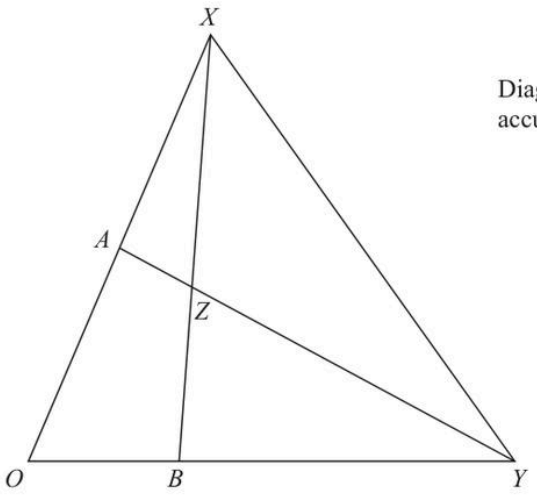


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows a triangle OXY

$\overrightarrow{OX} = 2\mathbf{a}$ and $\overrightarrow{OY} = 3\mathbf{b}$

A is the midpoint of OX and B is the point on OY such that $OB : BY = 1 : 2$
The lines XB and AY intersect at Z.

(a) Find $\overrightarrow{AB}$ as a simplified expression in terms of **a** and **b**

(1)

(b) Using a vector method, find $\overrightarrow{OZ}$ as a simplified expression in terms of **a** and **b**

(9)

The point M on XY is such that O, Z and M are collinear.

(c) Find $\overrightarrow{OM}$ as a simplified expression in terms of **a** and **b**

(3)

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P 6 2 2 8 3 A 0 3 4 4 0

19. Question 11 · part b · 9 marks · Working and answer

Working and answer

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20. Question 11 · part c · 3 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part c only.

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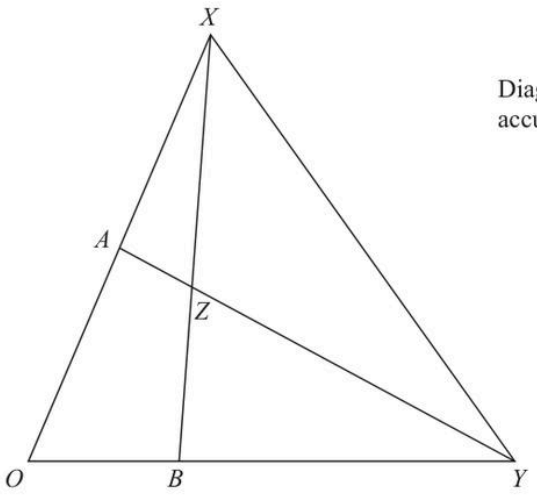


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows a triangle OXY

$$\overrightarrow{OX} = 2\mathbf{a} \text{ and } \overrightarrow{OY} = 3\mathbf{b}$$

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(a) Find $\overrightarrow{AB}$ as a simplified expression in terms of **a** and **b**

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(9)

The point M on XY is such that O, Z and M are collinear.

(c) Find $\overrightarrow{OM}$ as a simplified expression in terms of **a** and **b**

(3)

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20. Question 11 · part c · 3 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 11 continued

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20. Question 11 · part c · 3 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 11 continued

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20. Question 11 · part c · 3 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 11 continued

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(Total for Question 11 is 13 marks)

20. Question 11 · part c · 3 marks · Working and answer

Working and answer

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